ChargeGuard AI - Progress Report

PayPal AI Hackathon 2026 | Report date: 4 Nov 2026 | Submission deadline: 12 Nov 2026, 12:00 PT

1. What the project is

ChargeGuard is a two-sided tool for small merchants who sell subscriptions through PayPal. Customers describe a billing problem in plain words; the owner sees every problem as a *case* with an AI analysis, an evidence checklist and a draft reply, and approves it before anything touches PayPal. The goal is to resolve problems **before** they become disputes. Core rule: **the AI never touches money** - it proposes, a human approves, only then do we call PayPal.

2. Status at a glance

Plan phase	Status
Days 1-2: credentials, webhook receiver + signature verification, tunnel, seeder, schema	**DONE** - verified end to end
Days 3-4: case engine, rules layer, dispute ingestion, deadline clock	Next
Days 5-6: three AI jobs	Not started (AI client written, untested)
Days 7-8: owner dashboard, customer portal	Not started
Day 9-10: ledger, polish, video, README, deploy, submit	Not started

3. Decisions made

Decision	Choice	Why
Language / backend	TypeScript, Node 24, Fastify	One language across API and UI; shared types
Database	SQLite (better-sqlite3)	Zero setup; 7 tables fit easily
Validation	Zod	Same schema validates AI output; no free prose reaches the DB
AI provider	Any OpenAI-compatible endpoint (default Gemini)	No SDK lock-in; switch by changing 3 env vars
Frontend	React + Vite (planned)	Dashboard first, portal second
Tunnel	cloudflared quick tunnel	Already installed; free

4. What was built

File	Purpose
server/src/db.ts	All 7 tables with CHECK constraints on case and action statuses; storeEvent() = idempotent webhook storage keyed on paypal_event_id
server/src/paypal.ts	Cached OAuth token, paypal() fetch helper, verifyWebhook() calling PayPal's signature-verification endpoint
server/src/index.ts	POST /webhooks/paypal (verify, then store, then 200; bad or missing signature returns 400) and GET /health. Loads the repo-root .env

File	Purpose
server/src/ai.ts	generateJson(schema, system, user): one call for all three AI jobs, validates output with Zod
server/src/seed.ts	Creates a sandbox product, plan ($9.99/month) and subscription with custom_id; prints the buyer approval link
server/src/db.test.ts	One runnable check: duplicate event rejected, bogus action status refused
.env.example, .gitignore, LICENSE	Config template (no secrets), ignores .env / *.db / screenshots, MIT licence

5. How we verified it works

After the buyer approved subscription I-BN5BHFNLUH49 in the sandbox, PayPal reported it **ACTIVE** with a first $9.99 USD payment, and **8 signed webhooks** were stored in webhook_events, including **BILLING.SUBSCRIPTION.ACTIVATED** and **PAYMENT.SALE.COMPLETED**. Rows are stored only after PayPal's signature check passes, so this proves real verification works through the public tunnel. The Webhook Simulator could not be used for this: its mock events cannot be verified and are correctly rejected with 400.

6. Problems hit and how they were solved

Problem	Cause	Fix
Malformed Client ID in .env	Leading space after = in .env	Stripped; checked lengths and spaces without printing secrets
Server returned empty webhook_id	An older dev server (started before .env was filled) held stale env	Restarted it; server now reads the repo-root .env regardless of cwd
500 on unsigned requests	verify call threw on missing headers	Missing or invalid signature now returns a clean 400
Two .env files	Stray '.env.' typo copy (empty)	Deleted it
PayPal checkout: 'We're sorry' error	Sandbox accounts were set to India (recurring USD unsupported)	Created US Business + Personal accounts and a new REST app; new credentials and webhook
API keys in a screenshot	Credentials photographed in the project folder	Added Screenshot*.png to .gitignore; advised deleting and rotating

7. How to run it

```
cp .env.example .env        # fill PAYPAL_* and AI_API_KEY
cd server && npm install
npm run dev                 # API on :3000
cloudflared tunnel --url http://localhost:3000   # public URL for the webhook
npm run seed                # new sandbox subscription + approval link
npm test                    # DB idempotency check
```

The webhook URL in the PayPal dashboard must be `<tunnel-url>/webhooks/paypal`. A quick tunnel gets a new URL each restart, so update the webhook when that happens.

8. Open items and next steps

Item	Notes
Commit latest changes	ai.ts, seed.ts, dotenv path fix, 400 handling are not yet committed
Gemini API key	AI_API_KEY is still empty; needed from day 5
Transaction Search permission	Enable on the new US app now (up to 9 hours to take effect)
Case engine (days 3-4)	Webhook dispatcher into subscriptions/transactions tables handling out-of-order events; SQL rules for duplicate charge and failed payment; dispute ingestion (re-fetch seller_response_due_date); deadline clock
Buyer-side dispute setup	Needed for the dispute demo path; budget half a day
Hosting	Render for the final stable URL (replaces the quick tunnel); demo must stay live through 15 Dec 2026
Security housekeeping	Rotate the old India app secret; keep .env and screenshots out of the repo